

Metagenomics file formats, quality control and best practices in data management

Puspendu Sardar

University of Cambridge

2025.11.17

FASTA File

- Generally, `fasta` files contain reference genomes.
- They can also contain contigs and scaffolds – these are seen when making assemblies.
- We will have a look at some in the practical

Format of a FASTA definition line


The diagram illustrates the format of a FASTA definition line. It shows a sequence identifier followed by a description in square brackets, then the sequence name, and finally the sequence itself. Callouts indicate the placement of spaces and a hard return.
>Seq1 [organism=Carpodacus mexicanus] [clone=6b] actin (act) mRNA, partial CDS
CCTTTATCTAATCTTTGGAGCAYGAGCTGGCATAGTTGGAACCGCCCTCAGCCTCCTCATC

- You can count sequences with `grep`, but remember to be careful with `grep` here, always put “>”, not >

FASTQ File

- `fastq` files contain your raw sequence data (reads)
- The `fastq` format was originally developed at the Sanger Institute and released around the year 2000

FASTQ File

- `fastq` files contain:
 - The sequence identifier (and optionally a description), usually start with `@`
 - The raw sequence
 - The “+” symbol (optionally followed by the identifier/description)
 - The base quality scores are in the ASCII scale

```
@VH00558:32:AAAMF32HV:1:1101:5526:27750/1
GAAATGATTGCCTTTTCCAACAATCTGATCAACTGGAAACAGGAAATCATCAATTCA
+
-C;CCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCC;CCCCC;CCCCCCCCCCCCCCCC-CCCC
```

!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHI



$$Q = -10 \times \log_{10}$$

(Probability of Error)

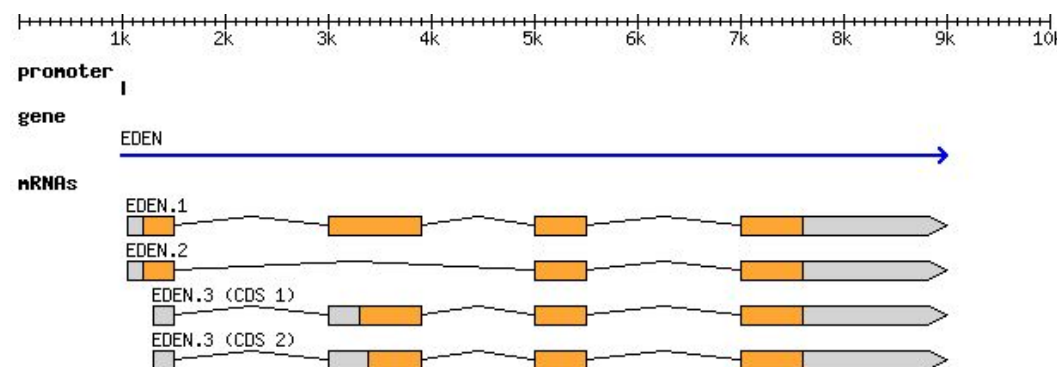
$$Q = 20 \rightarrow 10^{-2} = 0.01 = 1\%$$

$$Q = 30 \rightarrow 10^{-3} = 0.001 = 0.1\%$$

- `fastq` contains more detail than `fasta`

GFF File

Store coordinates of genomic regions



Full column specification [here](#)

```
##gff-version 3.1.26
##sequence-region ctg123 1 1497228
ctg123 . gene 1000 9000 . + . ID=gene00001;Name=EDEN
ctg123 . TF_binding_site 1000 1012 . + . ID=tfbs00001;Parent=gene00001
ctg123 . mRNA 1050 9000 . + . ID=mRNA00001;Parent=gene00001;Name=EDEN.1
ctg123 . mRNA 1050 9000 . + . ID=mRNA00002;Parent=gene00001;Name=EDEN.2
ctg123 . mRNA 1300 9000 . + . ID=mRNA00003;Parent=gene00001;Name=EDEN.3
ctg123 . exon 1300 1500 . + . ID=exon00001;Parent=mRNA00003
ctg123 . exon 1050 1500 . + . ID=exon00002;Parent=mRNA00001,mRNA00002
ctg123 . exon 3000 3902 . + . ID=exon00003;Parent=mRNA00001,mRNA00003
ctg123 . exon 5000 5500 . + . ID=exon00004;Parent=mRNA00001,mRNA00002,mRNA00003
ctg123 . exon 7000 9000 . + . ID=exon00005;Parent=mRNA00001,mRNA00002,mRNA00003
ctg123 . CDS 1201 1500 . + 0 ID=cds00001;Parent=mRNA00001;Name=edenprotein.1
ctg123 . CDS 3000 3902 . + 0 ID=cds00001;Parent=mRNA00001;Name=edenprotein.1
ctg123 . CDS 5000 5500 . + 0 ID=cds00001;Parent=mRNA00001;Name=edenprotein.1
ctg123 . CDS 7000 7600 . + 0 ID=cds00001;Parent=mRNA00001;Name=edenprotein.1
ctg123 . CDS 1201 1500 . + 0 ID=cds00002;Parent=mRNA00002;Name=edenprotein.2
ctg123 . CDS 5000 5500 . + 0 ID=cds00002;Parent=mRNA00002;Name=edenprotein.2
ctg123 . CDS 7000 7600 . + 0 ID=cds00002;Parent=mRNA00002;Name=edenprotein.2
ctg123 . CDS 3301 3902 . + 0 ID=cds00003;Parent=mRNA00003;Name=edenprotein.3
ctg123 . CDS 5000 5500 . + 1 ID=cds00003;Parent=mRNA00003;Name=edenprotein.3
ctg123 . CDS 7000 7600 . + 1 ID=cds00003;Parent=mRNA00003;Name=edenprotein.3
ctg123 . CDS 3391 3902 . + 0 ID=cds00004;Parent=mRNA00003;Name=edenprotein.4
ctg123 . CDS 5000 5500 . + 1 ID=cds00004;Parent=mRNA00003;Name=edenprotein.4
ctg123 . CDS 7000 7600 . + 1 ID=cds00004;Parent=mRNA00003;Name=edenprotein.4
```

SAM File

(Sequence Alignment Map)

- `sam` files contain information about how well your sequence data matches a reference genome (alignments)
- Let's have a look!

@HD VN:1.5 SO:coordinate											Header section
@SQ SN:ref LN:45											
r001	99	ref	7	30	8M2I4M1D3M	=	37	39	TTAGATAAAGGATACTG	*	Alignment section
r002	0	ref	9	30	3S6M1P1I4M	*	0	0	AAAAGATAAGGATA	*	
r003	0	ref	9	30	5S6M	*	0	0	GCCTAAGCTAA	* SA:Z:ref,29,-,6H5M,17,0;	
r004	0	ref	16	30	6M14N5M	*	0	0	ATAGCTTCAGC	*	
r003	2064	ref	29	17	6H5M	*	0	0	TAGGC	* SA:Z:ref,9,+,5S6M,30,1;	
r001	147	ref	37	30	9M	=	7	-39	CAGCGGCAT	* NM:i:1	

<

CIGAR Strings

(Concise Idiosyncratic Gapped Alignment Report)

CIGAR String: 3M1I3M1D5M

RefPos:	1	2	3	4	5	6	7		8	9	10	11	12	13	14	15	16	17	18	19
Reference:	C	C	A	T	A	C	T		G	A	A	C	T	G	A	C	T	A	A	C
Read:					A	C	T	A	G	A	A		T	G	G	C	T			

Op	BAM	Description
M	0	alignment match (can be a sequence match or mismatch)
I	1	insertion to the reference
D	2	deletion from the reference
N	3	skipped region from the reference
S	4	soft clipping (clipped sequences present in SEQ)
H	5	hard clipping (clipped sequences NOT present in SEQ)
P	6	padding (silent deletion from padded reference)
=	7	sequence match
X	8	sequence mismatch

FASTQ to SAM (Alignment)

- Build bowtie index from fasta file:

- `bowtie2-build your_fasta_file your_bowtie_index`

- Run bowtie mapping:

- `bowtie2 -x your_bowtie_index -1 read_1.fastq -2 read_2.fastq > read.sam`

BAM File

- `bam` files are compressed (binary) `sam` files.
- Generally, alignments are stored as `bam` files to save disk space

- Converting a `sam` file to a `bam` file:

```
samtools view -h demo.sam -o demo.bam
```

- Visualising `sam/bam` files:

```
samtools view -h your_file | less
```

- Or IGV for a graphical visualisation.

VCF File

- `vcf` files store variant calling information – they show the differences between your reads and the reference genome
- This is interesting because it can give insight into the genetic changes that may be driving disease
- Just like SAM -> BAM, binary VCF files are called BCF
- Some variant callers are: `bcftools` and `freebayes`

VCF File

Example

VCF header

```
##fileformat=VCFv4.0
##fileDate=20100707
##source=VCFtools
##reference=NCBI36
##INFO=<ID=AA,Number=1,Type=String,Description="Ancestral Allele">
##INFO=<ID=H2,Number=0,Type=Flag,Description="HapMap2 membership">
##FORMAT=<ID=GT,Number=1,Type=String,Description="Genotype">
##FORMAT=<ID=GQ,Number=1,Type=Integer,Description="Genotype Quality (phred score)">
##FORMAT=<ID=GL,Number=3,Type=Float,Description="Likelihoods for RR,RA,AA genotypes (R=ref,A=alt)">
##FORMAT=<ID=DP,Number=1,Type=Integer,Description="Read Depth">
##ALT=<ID=DEL,Description="Deletion">
##INFO=<ID=SVTYPE,Number=1,Type=String,Description="Type of structural variant">
##INFO=<ID=END,Number=1,Type=Integer,Description="End position of the variant">
```

Mandatory header lines

Optional header lines (meta-data about the annotations in the VCF body)

Body

#CHROM	POS	ID	REF	ALT	QUAL	FILTER	INFO	FORMAT	SAMPLE1	SAMPLE2
1	1	.	ACG	A,AT	.	PASS	.	GT:DP	1/2:13	0/0:29
1	2	rs1	C	T,CT	.	PASS	H2;AA=T	GT:GQ	0 1:100	2/2:70
1	5	.	A	G	.	PASS	.	GT:GQ	1 0:77	1/1:95
1	100	.	T		.	PASS	SVTYPE=DEL;END=300	GT:GQ:DP	1/1:12:3	0/0:20

Reference alleles (GT=0)

Alternate alleles (GT>0 is an index to the ALT column)

Deletion

SNP

Large SV

Insertion

Other event

Phased data (G and C above are on the same chromosome)

BAM to VCF

- `samtools sort file.bam -o file.sorted.bam`
- `freebayes -f reference.fasta file.sorted.bam > file.vcf`

Summary

- FASTQ → store (“raw”) sequencing data and quality scores for Illumina and base called ONT reads.
- FASTA → store DNA sequences, for example reference genomes or genome assemblies
- GFF/BED → tab-delimited file formats that store the coordinates of genomic features, such as gene annotations for a genome or primer locations.
- SAM/BAM → store alignment data, including sequence and quality information from the FASTQ file
- VCF → store genetic variance information based on the position of the reference genome
- CSV / TSV → comma-separated or tab-separated files storing tabular data, common output for many analysis software

Quality control and report

QC of raw sequencing data in fastq/fq/fastq.gz/fq.gz

Quality Control

- FASTQC (<https://www.bioinformatics.babraham.ac.uk/projects/fastqc/>) is your friend!
- Run FASTQC on any new sequencing data we have
- MultiQC (<https://multiqc.info/>)



A modular tool to aggregate results from bioinformatics analyses across many samples into a single report.

Report generated on 2024-02-12, 13:25 CET based on data in: /Users/ewels/GitHub/MultiQC/website/public/examples/bs-seq

don't show again 

General Statistics

- Violin plot

Showing $8/8$ rows and $10/15$ columns.

Sample Name	% mCpG	% mCHG	% mCHH	M C's	% Dups	% Aligned	% BP Trimmed
MethylC-Seq_mm_fc_1wk_SRR921767_1	77.0%	0.5%	0.5%	1 568.0	2.9%	86.0%	2.6%
MethylC-Seq_mm_fc_1wk_SRR921768_1	77.0%	0.5%	0.5%	1 559.9	2.9%	85.7%	2.9%
MethylC-Seq_mm_fc_1wk_SRR921769_1	77.0%	0.5%	0.5%	321.4	1.6%	85.7%	3.0%
MethylC-Seq_mm_fc_1wk_SRR921770_1	77.0%	0.5%	0.5%	1 562.8	2.9%	85.9%	2.8%
MethylC-Seq_mm_fc_2wk_SRR921694_1	75.7%	0.9%	1.1%	1 928.2	7.0%	78.4%	13.5%
MethylC-Seq_mm_fc_2wk_SRR921695_1	75.8%	0.9%	1.1%	1 930.1	7.0%	78.4%	13.6%

Installation and usage of `fastqc`

- `conda create -n fastqc`
- `conda activate fastqc`
- `conda install -c bioconda fastqc`
- `fastqc *.fastq`

Installation and usage of multiQC

- Configure conda channels:

- `conda config --add channels defaults`
- `conda config --add channels bioconda`
- `conda config --add channels conda-forge`
- `conda config --set channel_priority strict`

- `micromamba create -n multiQC`

- `micromamba activate multiQC`

- `micromamba install -c bioconda multiqc`

Best practices

Data handling, file naming, directory structure, and backup

Data handling

- **Document** what you have done
- Name files in an appropriate manner
- Keep a backup copy of important files (but don't keep too much redundancy)
- Generally best to compress things into binary to save disk space
- For example:
 - `bowtie2-build Acinetobacter_baumannii_XH386_v1.fa Acinetobacter_baumannii_XH386_v1-index`
 - `bowtie2 -x bowtie2-index -1 9119_545_1.fastq -2 9119_545_2.fastq > 9119_545.sam`
 - `samtools view -h 9119_545.sam -o 9119_545.bam`

File naming example

DC_20231009_DataManagement_FilesAndFolders.pdf

DC - a personal or lab code for different type of files

20231009 - Date in *yyyymmdd* or *ddmmyyyy* format, the most universally understood format, and is readable by most computing systems; alternatively a version number could be added here, e.g. V001, V002, ..., V010

DataManagement - Topic, meaningful and easy-to-read 'keyword' title, without punctuation or special characters

FilesAndFolders - Title, detailed 'keyword'

DO NOT use spaces and AVOID special characters whenever possible

PC_yyyymmdd_Topic_Filename_version.ending

PC - project code/number

version - NOT final_submission_edit_reallyfinal_...

Personal, design your own style but be **consistent**

Directory structure for scientific projects

```
./your-project-name/
├── 1_admin
│   ├── 1_proposals    <-- project background, plans, proposals and applications...
│   ├── 2_finance      <-- funding applications, receipts, project budgets...
│   └── 3_reports
├── 2_ethics
│   ├── 1_approval     <-- ethics application, approval docs...
│   └── 2_consent       <-- consent forms, questionnaires...
├── 3_experiment
│   ├── 1_collection   <-- data collection scripts, experimental stimuli and code...
│   ├── 2_data
│   │   ├── 1_raw      <-- original (read only) data files, e.g. raw reaction time data, etc...
│   │   ├── 2_interim  <-- intermediary files, pre-processed or transformed data...
│   │   └── 3_processed <-- final canonical datasets for modeling and stats...
│   ├── 3_analysis(*)
│   │   ├── funcs      <-- source functions for processing, modeling, visualisation....
│   │   ├── models     <-- trained models, model predictions, coefficients...
│   │   └── notebooks  <-- Jupyter notebooks for data exploration, plotting...
│   └── 4_outputs
│       ├── 1_figures   <-- generated figures and graphics for reporting...
│       └── 2_reports   <-- project reports, HTML, PDF...
└── 4_dissemination
    ├── 1_presentations <-- Keynote/PowerPoint slides, conference posters...
    ├── 2_publications  <-- journal articles, published conference proceedings...
    └── 3_publicity     <-- media coverage and publicity, newspaper articles, outreach...
```


Data storage and backup

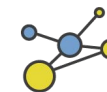
- Store important files in a central shared location (whenever possible)
- Make sure important files are backed up
E.g., reads (~~fastq/fq/fq.gz~~/fastq.gz files) and reference genomes (~~fasta/fa/fa.gz~~/fasta.gz files)
- It is good to have a backup of important files on a different file storage system and/or physical location if possible



Findable



Accessible



Interoperable



Reusable

"Data doesn't really exist unless you have at least two copies of it."

— [Jack Schofield](#) (2008)

Thanks for your attention!

References

- Li, H. *et al.* (2009) “The sequence alignment/map format and SAMtools,” *Bioinformatics*, 25(16), pp. 2078–2079.
<https://doi.org/10.1093/bioinformatics/btp352>
- Sequence Alignment/Map Format Specification.
<https://samtools.github.io/hts-specs/SAMv1.pdf>
- Langmead, B. and Salzberg, S.L. (2012) “Fast gapped-read alignment with bowtie 2,” *Nature Methods*, 9(4), pp. 357–359.
<https://doi.org/10.1038/nmeth.1923>
- Andrews S. (2010). FastQC: a quality control tool for high throughput sequence data. <http://www.bioinformatics.babraham.ac.uk/projects/fastqc>
- Garrison E, Marth G (2012). Haplotype-based variant detection from short-read sequencing. *arXiv preprint arXiv:1207.3907 [q-bio.GN]*